



Effect of band gap of graphene oxide on interaction with bovine serum albumin: Correlation of band gap with sensitivity

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ABSTRACT

In the direction of developing new and reliable prognostic tools, metamaterial-based biosensor device is an emergent potential field. The special design of metamaterial provides signature frequency features of healthy or diseased tissues. In the current work we have studied the single split resonance device (SSR) using graphene oxide (GO) as probe for the accelerated and selective detection of bovine serum albumin (BSA). Further, we have used different band gap GO to interpret the structure-property relationship. Spectroscopy methods have been employed to further support the SSR results. 1 ppm solution of mGO-0.5 (band gap ~ 1.35 eV) shows sequential increase in sensitivity on increasing amount of BSA. Figure-of-merit (FoM) of SSR biosensor also increased from 57 to 82 and highest Q-factor calculated for mGO-1 is ~ 47 making it a good candidate as probe molecule. This work lays out a correlation of degree of GO functionalization with a physical parameter, i.e., band gap for activity towards protein molecules. Also, as the SSR devices use large wavelength radiation with high resolution and accuracy for quantitative estimation of analyte, these are potential non-invasive point-of-use portable diagnostic and early disease diagnosis devices.

1. Introduction

Metamaterials based biosensors use the high frequency electromagnetic waves (MHz to GHz), efficient enough to interact with most of the materials, causing change in its electromagnetic response [1,2,3]. This response is studied as the change in materials reflection / transmission coefficient due to the change in the permeability and permittivity parameters [4]. Islam et al., designed the triple circle split ring resonator for the recognition of various oils, fluids etc. [5]. Tobias et al. [6], reported single split resonator (SSR) biosensor device for the aptamer-based detection of C-reactive proteins. Lee et al. [7], reported a planar SSR device for recognition of prostate specific antigen (PSA) and cortisol stress hormone. In search of potential probe molecules for split ring resonator biosensor device, we have recently reported the quantitative detection of BSA using perylenediimide (PDI) as probe molecule [8]. Probe molecules play important role for quantitative detection as they form charge-transfer complex with the analyte and deliver special finger-point frequency signals with increasing amount of analyte due to increased concentration of C-T complex affecting the dielectric constant

of the medium. Zhang et al., used silver nanoparticles decorated molybdenum disulfide (MoS_2) as probe for detection of biogenic amines [9]. Probe molecules accelerate the response due to effective change in dielectric constant of the medium resulting in the frequency shift.

Graphite, graphene oxide (GO), and reduced graphene oxide (rGO) are promising nanomaterials for biomedical applications [10,11,12,13,14]. Studies have been performed on the nanoparticle-decorated reduced graphene oxide with BSA in bone tissue engineering [15]. It is also observed that BSA attachment on GO increases the antimicrobial activity and hemocompatibility [16]. There are only a few studies on GO interaction with proteins. Bapli et al., studied the effect of graphene oxide on the interaction between human serum albumin (HSA) protein and hydrophilic fluorescent dye analysed by spectroscopic methods [17]. GO possesses special properties due to the functionalities present on the basal plane as well as opto-electronic properties to be used in biosensors [18,19]. Disease detection with precision is performed by modifying the GO surface with functionalities to bind with specific biomarkers [20,21]. Degree of oxidation of GO is one of the important parameters to tune the sensitivity towards respective analyte due to

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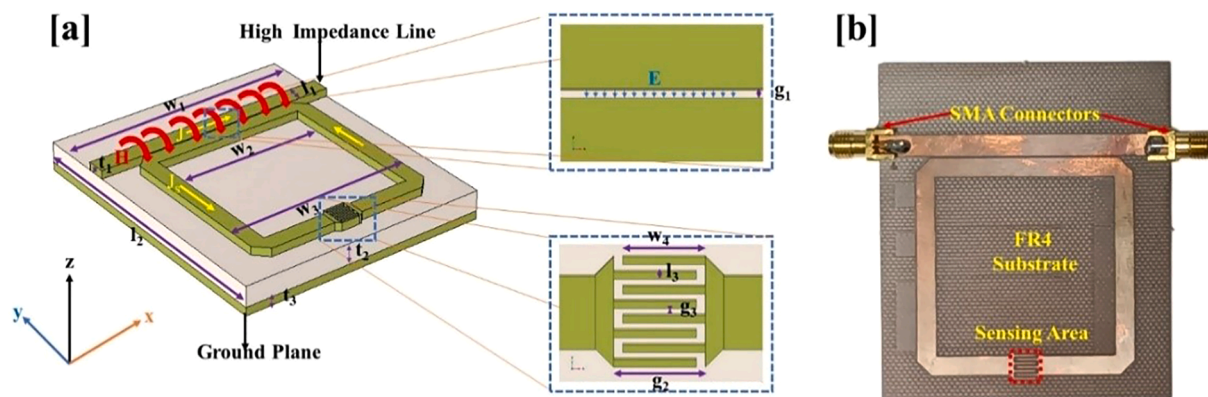


Fig. 1. Illustration of numerically simulated and experimental fabricated SSR-based biosensor. (a) The schematic representation of biosensor device consists of SSR with a high impedance microstrip line of thickness $t_1 \sim 35\mu\text{m}$, length $l_1 = 4.5\text{mm}$, and width $w_1 = 60\text{mm}$. The microstrip line is far apart from SSR through the gap $g_1 = 0.3\text{mm}$ and it also consists of a ground plane of thickness $t_3 \sim 35\mu\text{m}$ and dielectric substrate of permittivity $\epsilon \approx 3.38$, thickness $t_2 \sim 1.524\text{mm}$, and length $l_2 = 70\text{mm}$. The propose SSR is rectangular in shape and inner and outer width is $w_2 = 37\text{mm}$ and $w_3 = 45\text{mm}$ respectively, so its resultant width is 4mm . The split area ($g_2 = 5\text{mm}$) of SSR consists of eight fingers of width $w_4 = 4.5\text{mm}$, length $l_3 = 0.4\text{mm}$, and are having gap $g_3 = 0.4\text{mm}$ from each other. The time varying magnetic field is used for current mode in propose SSR. (b) The photograph depicts a fabricated sample with two SMA connectors and a dotted rectangle shows the sensing area.

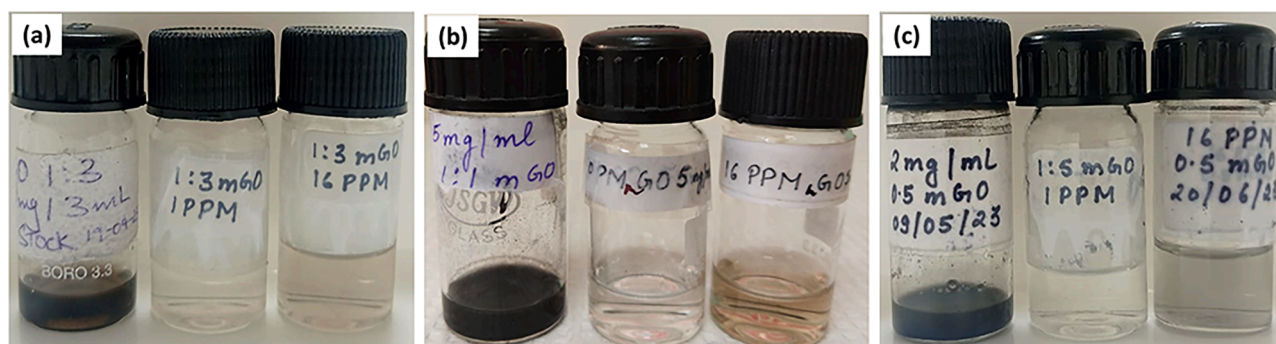


Fig. 2. Pictures of stock, 1 ppm and 16 ppm solutions of (a) mGO, (b) mGO-1, and (c) mGO-0.5.

band gap tuning in semi-conductor range.

For biological applications we need reproducible, direct method of graphene oxide preparation with specific degree of oxidation without involving toxic chemicals and multistep synthesis process so that its bioactivity of GO does not change from batch to batch and lab to lab. Degree of oxidation controls the functionalization as well as the sp^2 lattice size of GO which further influences the sensitivity of GO based biosensors [22]. There is no study performed yet for the bioactivity of graphene oxide with respect to band gap of GO which is directly tuned by degree of oxidation. The current work focuses on the selective detection of BSA analyte with graphene oxide synthesized by our established mild oxidation process, i.e., mGO, mGO-1 & mGO-0.5 with 2.2 eV, 1.5 eV, and 1.35 eV band gap respectively as probe molecule to correlate the structure property relationship [23,24,25]. The sensitivity of these graphene oxides towards BSA is calculated using absorption spectroscopy as well as by fabricating single split resonator (SSR) biosensor devices. To the best of our knowledge, this is the first report where, GO band gap is correlated with its activity towards sensing BSA protein using spectroscopy method as well as frequency based SSR device.

2. Experimental section

2.1. Materials and methods

Bovine serum albumin was purchased from Thermo Scientific, USA (crystalline, 98 %). The ovalbumin (Grade V minimum 98 %, crystallized and lyophilized) was purchased from Sigma (St. Louis, MO, USA).

Solutions were prepared with deionized (DI) water processed with a MilliQ ultra-high purity water system (Millipore, Bedford, MA, USA). All other chemicals used in this study were of analytical grade. GO was synthesized and characterized as per our previously reported methods and found same [23–25]. UV–Vis. spectroscopy measurements were performed on LabIndia UV–Vis., spectrophotometer.

2.2. Single split resonator device fabrication and measurement

Fig. 1(a) depicts the proposed design of a biosensing device intended for the detection of various concentrations of BSA in DI (de-ionized) water using GO solutions. The proposed SSR consists of a rectangular structure with a split and a microstrip transmission line and is a RLC series resonant circuit. When an electromagnetic wave meets SSR's resonant frequency, the electromagnetic wave creates a standing wave that leads to strong electric fields inside the split. In Fig. 1(a), SSR is rectangular in shape with truncated edges and a single split. The proposed design topology of SSR effectively prevents reflection at the edges and produces high current density (J_s) on the surface of structure. The device comprises a copper-coated SSR and a planar transmission microstrip line, which are employed for the transmission of electromagnetic radiation. The spacing between them is 0.3mm , and this spacing has been calibrated to ensure the optimum possible coupling of incident radiation for the frequency range that is being employed. The microstrip line is designed for an impedance of 50Ω and it is used to excite the resonator as electromagnetic waves couple with the SSR,

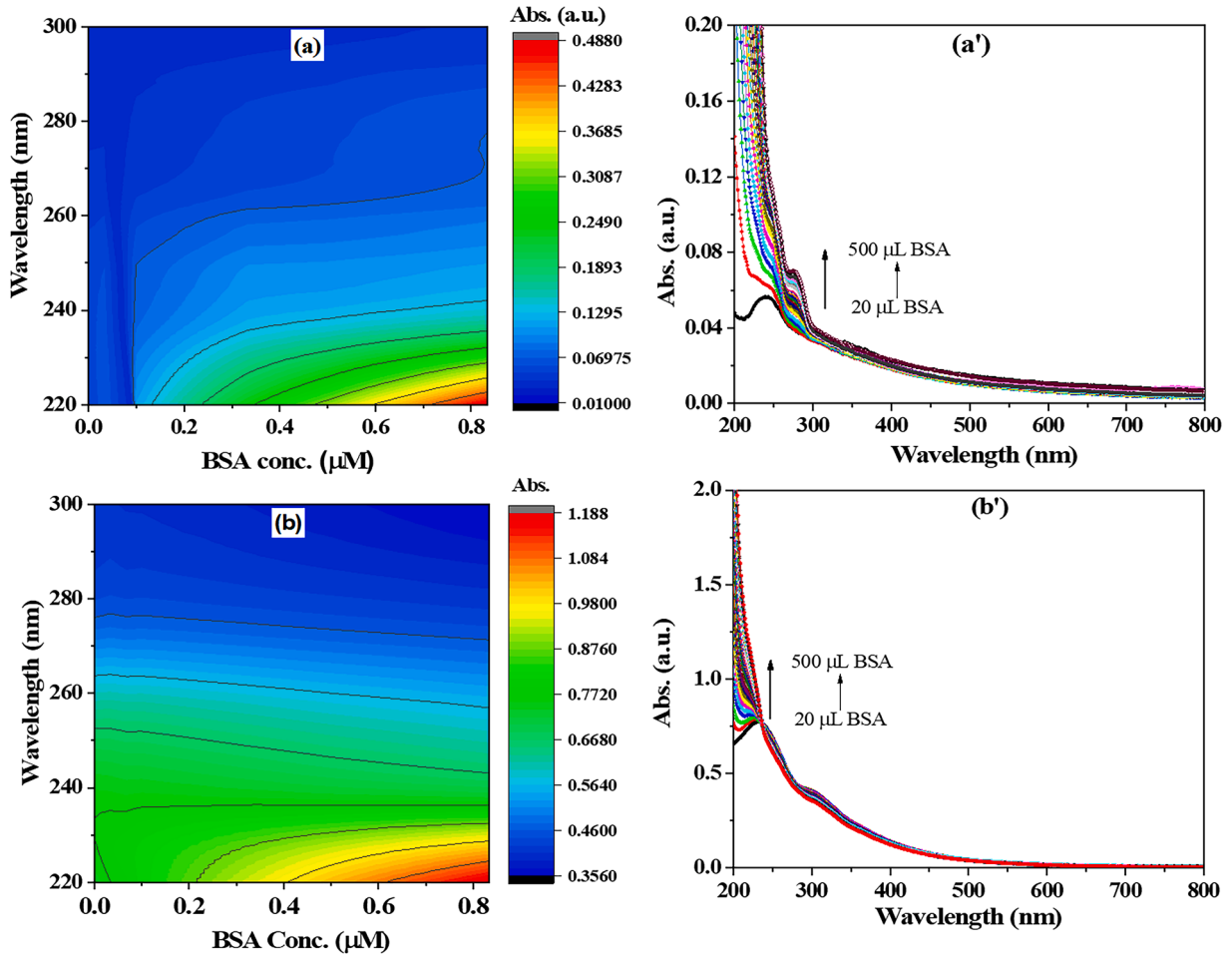


Fig. 3. Contour plot of mGO (a & b) with absorption spectra (a' & b') for 1 ppm and 16 ppm solutions with BSA titration.

Table 1

Molar absorptivity, binding constant, and % sensitivity for the graphene oxide solutions towards BSA.

	ϵ_c ($\times 10^6$ M ⁻¹)	K_c ($\times 10^6$)	%S _{abs} @ 0.833 μM
mGO (1 ppm)	2.2	13.3	108
mGO (16 ppm)	1.48	12.5	41.6
mGO-1 (1 ppm)	2.0	4.9	70.8
mGO-1 (16 ppm)	1.3	32.6	21.4
mGO-0.5 (1 ppm)	1.1	93	81.4
mGO-0.5 (16 ppm)	1.1	57	78.2

ϵ_c = absorption coefficient; K_c = binding constant; %S = % sensitivity.

resulting in a strong current that amplifies the time-varying magnetic field. The zoom vision of the split area in Fig. 1(a) illustrates a construction of the resistive type that has eight fingers that are 0.4mm in length and 4.5mm in width. Each finger is separated from the other by 0.4mm distance. These eight resistive electrodes are accountable for the penetration depth of the electric field into the sample due to strong coupling of electromagnetic waves. The device was calibrated using calibration module before run on Agilent vector network analyzer (VNA). The device was connected to VNA using subminiature version A (SMA) connector.

The input impedance $Z_{in}(\omega)$ from the microstrip transmission line of propose SSR is frequency independent and depends upon real and imaginary value of resistance R and $X(\omega)$. It can be modelled with the following equation

$$Z_{in}(\omega) = R + j\left(\omega L - \frac{1}{\omega C}\right) = R + jX(\omega) \quad (1)$$

where L and C are the inductance and capacitance of the structural elements, respectively. And ω is the angular frequency. The resonant frequency can be written as

$$f_0 = \frac{\omega_0}{2\pi} = \frac{1}{2\pi\sqrt{LC}} \propto \frac{1}{\sqrt{\epsilon_{eff}}} \quad (2)$$

where, ϵ_{eff} is the effective permittivity. In order to make a prediction regarding the resonant frequency and the surface current distribution of the resonator [7,26]. We have employed COMSOL Multiphysics 5.4 for electromagnetic simulation. Fig. 1(b) shows the fabricated sample of propose SSR with SMA connectors. It is fabricated with the help of MITS PCB prototyping system on double side copper (Cu) coated FR4 substrate.

The SSR is characterized via a reflection measurement (S_{11}), i.e. the amplitude ratio of the reflected wave voltage (V_1^+) to the incident wave voltage (V_1^-). The S_{11} amplitude depends on the frequency for the depicted resonator with air environment. Hereby, the resonance frequency is found at the maximum S_{11} value. The magnitude of S_{11} coefficient is defined as

$$S_{11} = 20\log_{10}\left(\frac{V_1^-}{V_1^+}\right) \quad (3)$$

The device sensitivity in MHz/DCU (megahertz per dielectric constant unit), quality factor (Q-factor) and Figure of merit (FoM) is

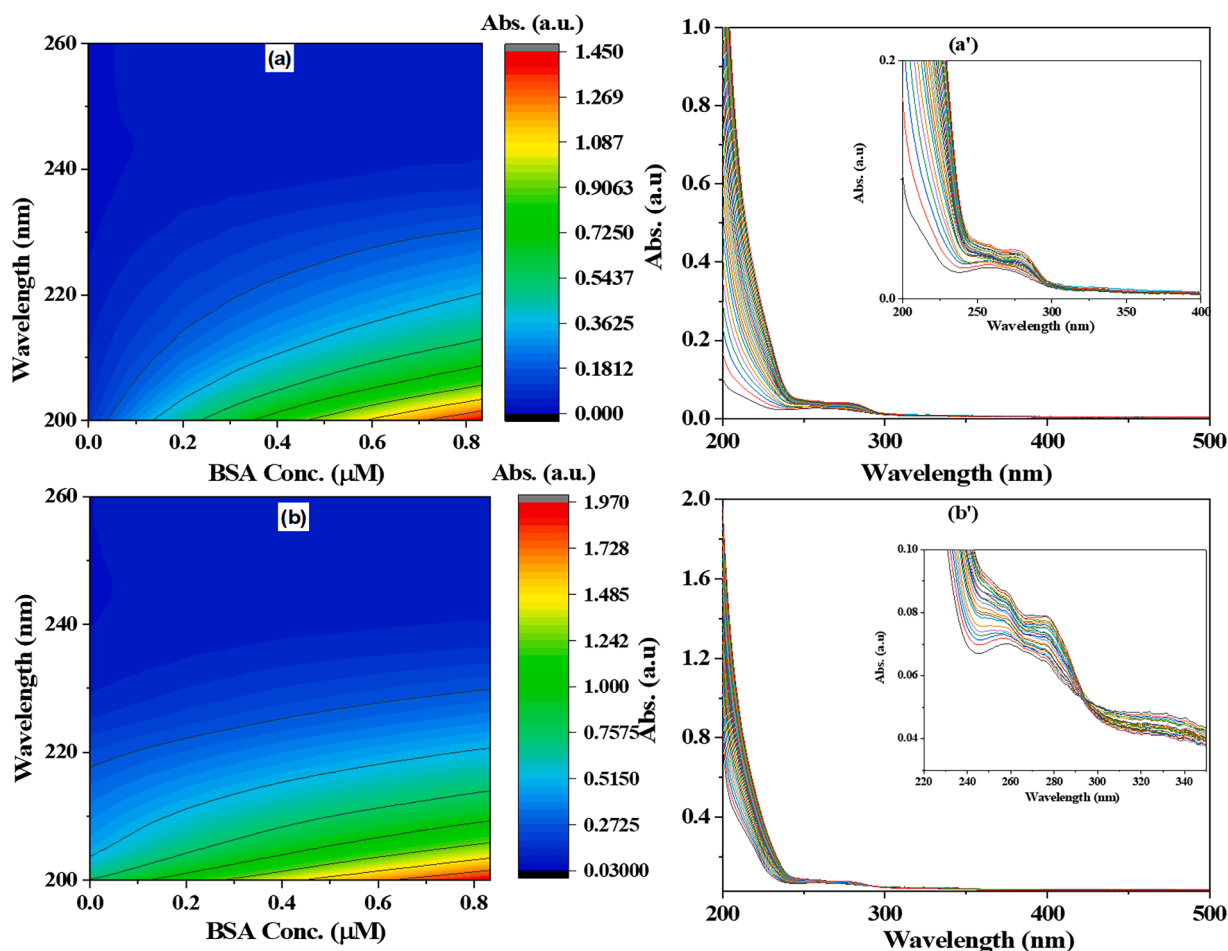


Fig. 4. Contour plot of mGO-1 (a & b) with absorption spectra (a' & b') for 1 ppm and 16 ppm solutions with BSA titration.

calculated with reference to air using following equations (4), (5) and (6)

$$S = \left| \frac{\Delta f_R}{\Delta \varepsilon} \right| = \left| \frac{f_{R1} - f_{R2}}{\varepsilon_1 - \varepsilon_2} \right| \quad (4)$$

$$Q \text{ factor} = \frac{f_R}{\text{FWHM}} \quad (5)$$

$$\text{FoM} = S \times Q \quad (6)$$

2.3. Preparation of graphene oxide (GO) and BSA solution (Scheme S1, supporting information)

GO of different band gap is synthesized by optimising the KMnO₄: H₂SO₄ ratio and characterized as per our reported methods [23–25]. The FTIR spectra of thus synthesized mGO, mGO-1 and mGO-0.5 is shown in SI (Figure S1A). By dissolving 100 mg of graphene oxide in 20 ml of distilled water (DW), the stock solution of graphene oxide was prepared followed by ultrasonication to exfoliate the GO sheets. Graphene oxide working solutions of different concentrations (1 ppm & 16 ppm) were prepared in DW to perform the interaction study (Scheme S1, supporting information). 5 μM BSA solution was prepared in 10 ml DW and homogenized thoroughly. The solutions were freshly prepared before experiment.

2.4. Benesi-Hildbrand equation and % sensitivity calculation

The interaction of GO with BSA can be expressed in terms of binding constant by fitting absorption data in Benesi-Hildbrand Eq. (7) [27],

$$\frac{A_o}{A_c} = \frac{1}{\varepsilon_c} + \frac{1}{\varepsilon_c \times K_c} \times \frac{1}{D_o} \quad (7)$$

where, A_o and A_c are the absorbance of GO in absence and presence of BSA. ε_c is the molar absorptivity of the GO-BSA complex, K_c is the binding constant and D_o is the concentration of BSA and where $[A_o] \gg [D_o]$.

The % sensitivity (% S) can also be estimated by absorption interaction data using the change in absorption to establish the sensitivity of GO for BSA. The equation 8 is used for calculation of % S

$$\%S = \frac{A_o - A_c}{A_o} \times 100 \quad (8)$$

3. Results and discussion

The most abundant protein in blood serum is serum albumin and due to structural model similarity, bovine serum albumin (BSA) is commonly used protein to study the effects of interaction with different molecules like organic and nano drug carriers, drug molecules, probes, etc. [28,29,30]. We have taken BSA as the model protein and studied the interaction of mGO, mGO-1 and mGO-0.5 (1 ppm and 16 ppm solution) with 5 μM BSA solution using absorption spectroscopy to completely understand the mechanism of interaction and band gap effect on sensitivity, Q-factor and figure-of-merit of the SSR device. 1 ppm and 16 ppm solutions of GO are clear and stable as shown in Fig. 2. For comparison, GO synthesized by conventional Hummer's process (HGO) with band gap of 3.4 eV was also studied for interaction with BSA (Figure S1B).

Ground state absorption spectroscopy is an important tool for GO

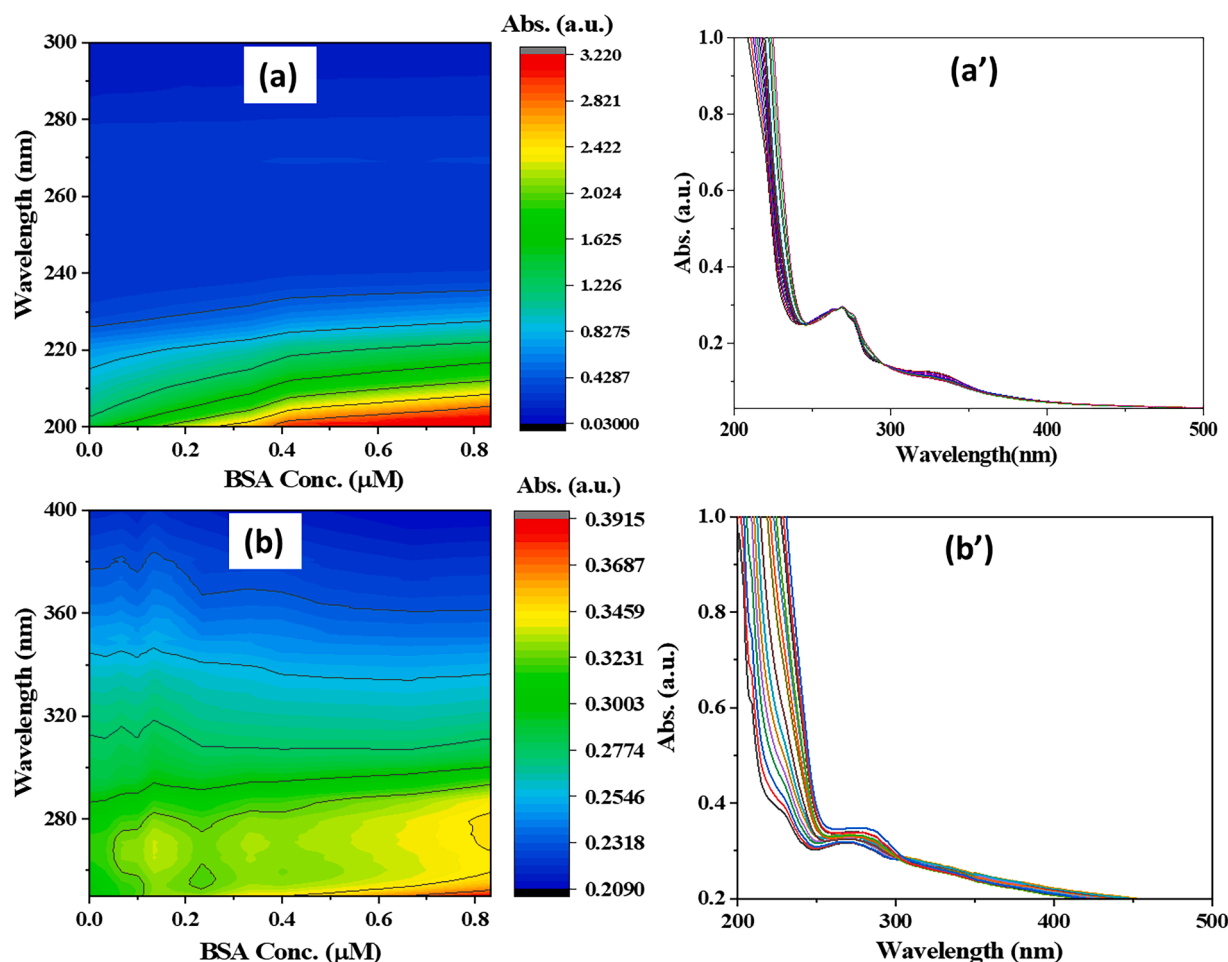


Fig. 5. Contour plot of mGO-0.5 (a & b) with absorption spectra (a' & b') for 1 ppm and 16 ppm solutions with BSA titration.

characterization to estimate the band gap which depends on the degree of oxidation. mGO shows the absorption maxima ~ 234 nm while, on controlled oxidation, the λ_{max} red shifts and observed at 266 nm in mGO-1 and further to 277 nm in mGO-0.5 [25]. The band gap is calculated using Tauc plot and found to be 1.8 eV, 1.5 and 1.35 eV on lowering the t-graphite: KMnO_4 ratio from 1:3 to 1:1 and 1:0.5 for mGO, mGO-1 and mGO-0.5 (Figure S1) [23–25]. The controlled and mild oxidation process for synthesis, results in low functionalized and highly conducting graphene oxide with least defects. As reported earlier by us, [23–25] mGO and mGO-1 show sp^2 cluster size of ~ 4.6 nm and 6.6 nm for mGO-0.5 confirming the long-range ordering imparting black color to these graphene oxide solutions. Absorption spectroscopy plays important role in studying the interaction of probe molecules with biomolecules and allows to understand the type of interaction based on absorption shifts. The surface of GO is negatively charged and acts as donor as also observed by us previously where GO acted as donor in presence of acceptor molecules [20]. The optical band gap is directly correlated with the sp^2 cluster size based on the Robertson correlation [31] and therefore, larger number of six-fold rings are present in mGO, mGO-1 and mGO-0.5 as in rGO. The XPS analysis showed that C/O ratio also increased from 0.8 in mGO to 1.09 in mGO-1 and 1.7 in mGO-0.5 indicating a lower degree of oxidation in mGO-1 and further in mGO-0.5 [23–25]. This gives sufficient surface to interact with BSA molecules of the size of ~ 7 nm making them ideal substrate for immobilization to be used as probe molecule [19,20,32].

3.1. Interaction study of mGO with BSA using absorption spectroscopy

The interaction with BSA is performed in aqueous solution (neutral) using absorption spectroscopy to avoid any interference and competition of solvent molecules and ideal behaviour could be understood. Pure BSA shows two absorption bands, i.e., at 210 nm for $n \rightarrow \pi^*$ transition of polypeptide backbone and at 278 nm for $\pi \rightarrow \pi^*$ of Tyr, Trp and Phe in BSA facilitating the ground state complex formation (Figure S2). As shown in Fig. 3a & 3a', on addition of BSA, the absorption band for $\pi \rightarrow \pi^*$ transition of 1 ppm mGO at 234 nm is red shifted to 249 nm with increase in absorbance (hyperchromic shift). New absorption band for BSA appears at 278 nm and strengthens with increasing amount of BSA without any shift. Redshift in absorption of dye was previously observed on addition of HSA and in ternary system along with GO due to formation of hydrogen bonding between host and guest molecules resulting in stabilization of excited state [17]. The redshift in absorption maxima of mGO indicates effective hydrogen-bonding interactions between mGO and BSA functional groups facilitating charge-transfer interactions (Fig. 3a). For 16 ppm mGO, the behaviour was completely different on titration with BSA (Fig. 3b, 3a'), where, absorbance of BSA dominated with appearance of isosbestic point at ~ 238 nm. The new species shows strong absorbance below 232 nm. 16 ppm mGO solution contains higher amount of graphene oxide sheets in close proximity due to strong π - π interactions between the sheets and therefore allows lesser surface for BSA molecules to interact with GO functional groups and weak ground state complex formation at higher GO concentration.

Binding constant is an important parameter to understand the molecular interactions and affinity to bind. Benesi-Hildebrand equation

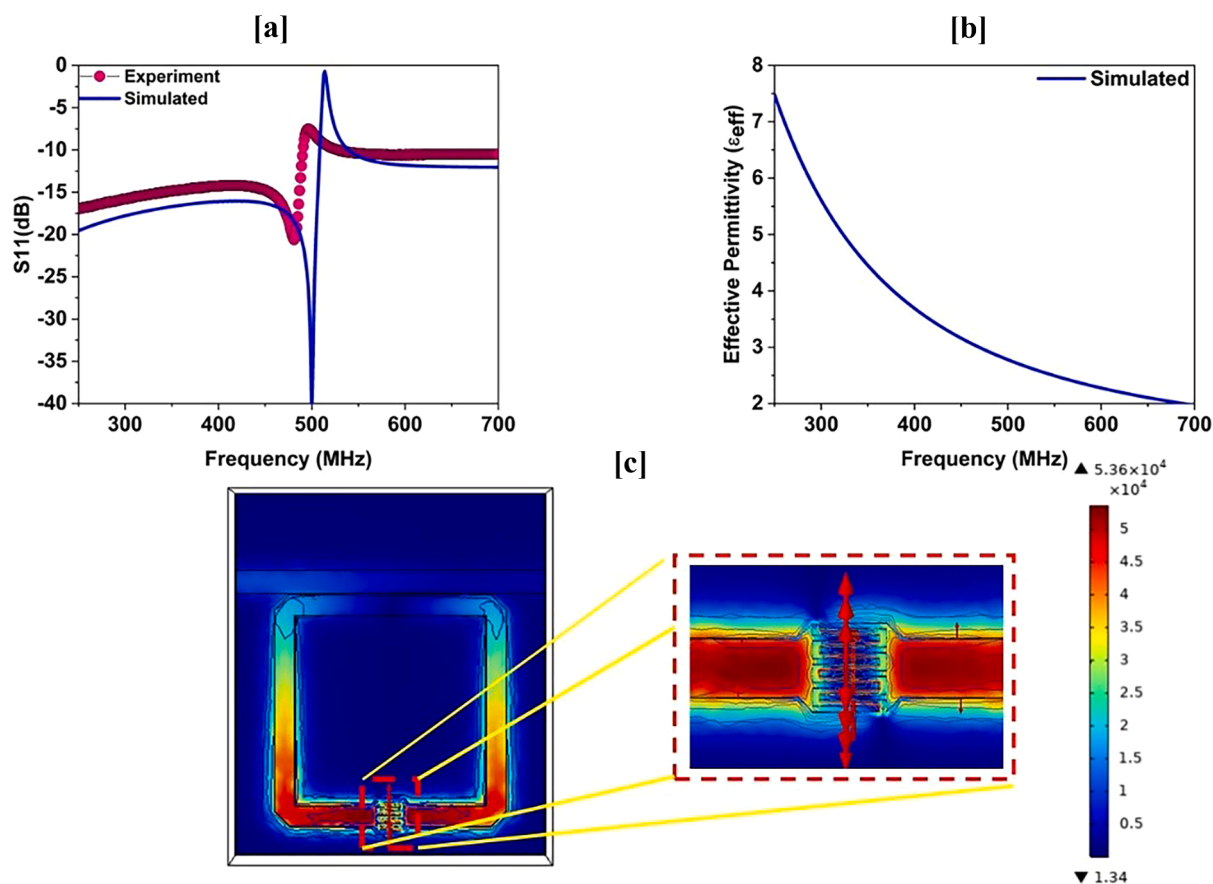


Fig. 6. (a) Experimental measured and numerically simulated resonant frequency (f_r) at 481 MHz and 500 MHz, (b) Numerically calculated effective permittivity (ϵ_{eff}) of simulated SSR device, (c) Electric field profile (E) of simulated SSR device. The zoom extent vision shows the electric field vectors' coupling at the split area.

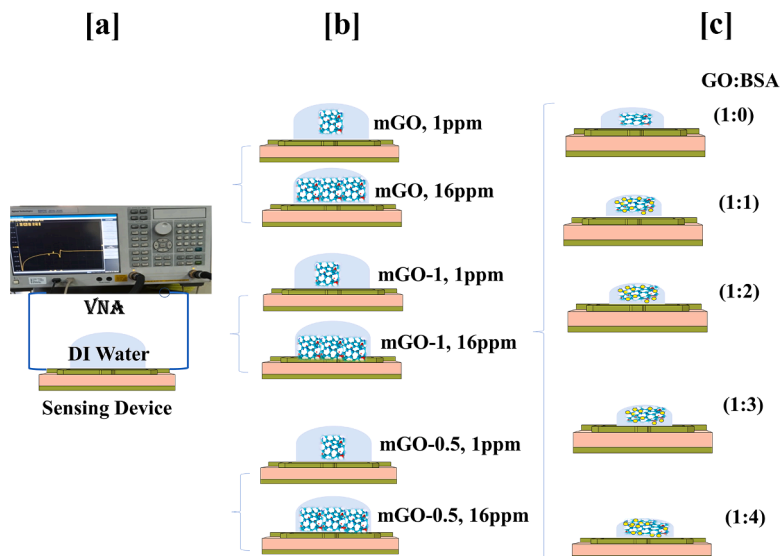


Fig. 7. Schematic illustration of the proposed device's sensing procedure. (b), (c), and (d) illustrate the diverse methodologies employed to detect synthesis compounds at varying concentrations.

(Eq. (7)) is used to determine the properties of complexes formed by charge-transfer interaction and hydrogen bonding (Figure S3, S4) [33, 34]. The calculated molar absorptivity and binding constant are summarized in Table 1. In case of mGO with band gap ~ 2.2 eV, 1 ppm solution gives the higher value of molar absorptivity and binding affinity

compared to 16 ppm solution. The sensitivity is also higher for 1 ppm mGO solution (~ 108 %).

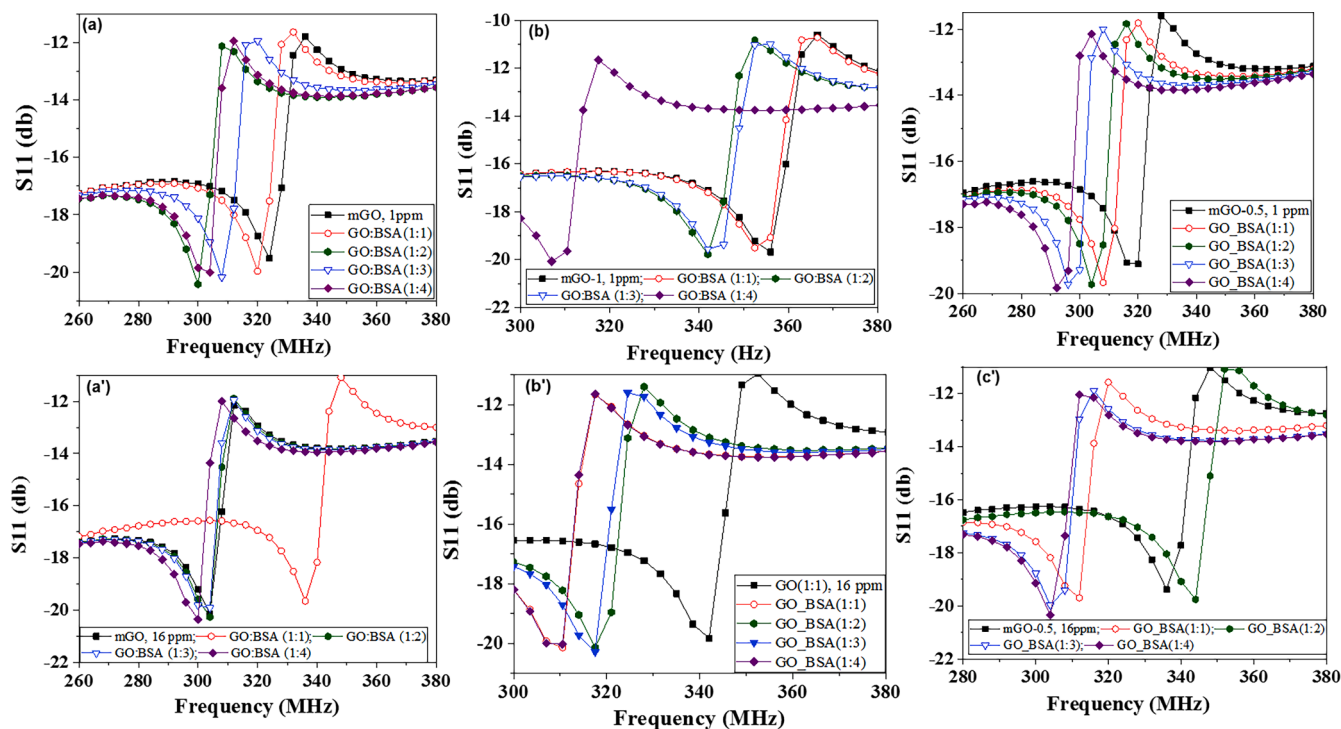


Fig. 8. Frequency shift of pure GO solution is employed as a solvent and in the presence of varying concentrations of BSA in (a) mGO 1 ppm solution, (a') mGO 16 ppm solution, (b) mGO-1 1 ppm solution, (b') mGO-1 16 ppm solution, (c) mGO-0.5 1 ppm solution, and (c') mGO-0.5 16 ppm solution.

Table 2

The experimental measurement of the resonant frequency, as well as the theoretical calculation of sensitivity, Q-factor, and figure of merit of the fabricated SSR in the presence of different concentrations of synthesis mGO-BSA.

GO:BSA	mGO				mGO-1				mGO-0.5			
	f_R (MHz)	Q-factor	S MHz/DCU	FoM	f_R (MHz)	Q-factor	S MHz/DCU	FoM	f_R (MHz)	Q-factor	S MHz/DCU	FoM
1:0 1 ppm	324	46	1.76	81	356	34	1.69	57.6	320	31	1.76	54.6
16 ppm	304	46.6	1.75	81.5	342	40	1.74	69.6	336	40	1.75	70
1:1 1 ppm	320	48.5	1.76	85.5	352.5	33	1.7	56.4	308	43	1.75	75.4
16 ppm	336	47.2	1.75	82.6	310.5	48	1.75	84.3	312	46	1.75	80.9
1:2 1 ppm	299.8	52.6	1.74	91.5	342	35	1.74	60.9	304	45	1.75	75.4
16 ppm	304	46.6	1.75	81.5	317.5	38	1.76	67	344	41	1.73	71.1
1:3 1 ppm	307.8	48	1.75	84.1	342	36	1.74	62.6	296	40	1.73	69.3
16 ppm	307	39.5	1.75	69.2	317	43	1.76	75.8	304	36	1.74	62.9
1:4 1 ppm	304	37	1.74	64.7	307	47	1.75	82.4	292	39	1.72	67.1
16 ppm	300	44	1.74	76.6	310.5	46	1.75	80.8	304	46	1.74	80.4

3.2. Interaction study of mGO-1 & mGO-0.5 with BSA using absorption spectroscopy

To further understand the interaction behaviour of BSA, lower band gap GO, i.e., mGO-1 and mGO-0.5 are studied for interaction and absorption spectra with titration of BSA is presented in Figs. 4 and 5 respectively. The absorbance is increased in range of 200–300 nm for both 1 ppm and 16 ppm mGO-1 solutions and no shift in π - π^* transition was observed on addition of BSA. Both the solutions of mGO-1 showed isosbestic point at 294 nm suggesting formation of interacting species with BSA and presence of bound and free BSA in equilibrium. mGO-0.5 with band gap of 1.35 eV also shows increase in absorbance for π to π^* transitions with addition of BSA solution and isosbestic point appears at 295 nm at both concentrations (Fig. 5). The absorbance for n to π^* transition decreases which indicating graphene oxide interaction with BSA molecules via non-bonding electrons of functional groups present on GO basal plane or edges through H-bonding. The molar absorptivity decreases as compared to mGO for both the mGO-1 and mGO-0.5 solutions (1 ppm and 16 ppm). Binding constant (K_b) is highly increased for the interaction of BSA with mGO-0.5. Presence of functional groups

and sp^2 lattice size play important role in sensitivity towards BSA molecules. 1 ppm solution shows better sensitivity for all the GO types. We also studied the interaction of HGO, which is highly oxidized graphene oxide, with BSA and spectra are shown in Figure S5. HGO did not show noticeable change in absorption on addition of BSA. Among all, mGO-0.5 (1 ppm) shows the highest binding constant value and sensitivity ~81 %.

To further understand the role of band gap and degree of functionalization on the sensitivity towards BSA, we fabricated SSR biosensor devices to study the shift in resonance frequency due to permittivity changes with higher accuracy and rapid detection. In the current study we are using GO with different band gap as probe targeting quantitative detection of BSA with high sensitivity and figure of merit. Probe molecules are sensitive and selective towards the analytes and enhance the response of the device. As GO sheets provide planar surface for BSA molecules and polar functional groups for charge-transfer interactions, SSR study is useful to understand the type of GO with most suitable band gap for detection of BSA protein.

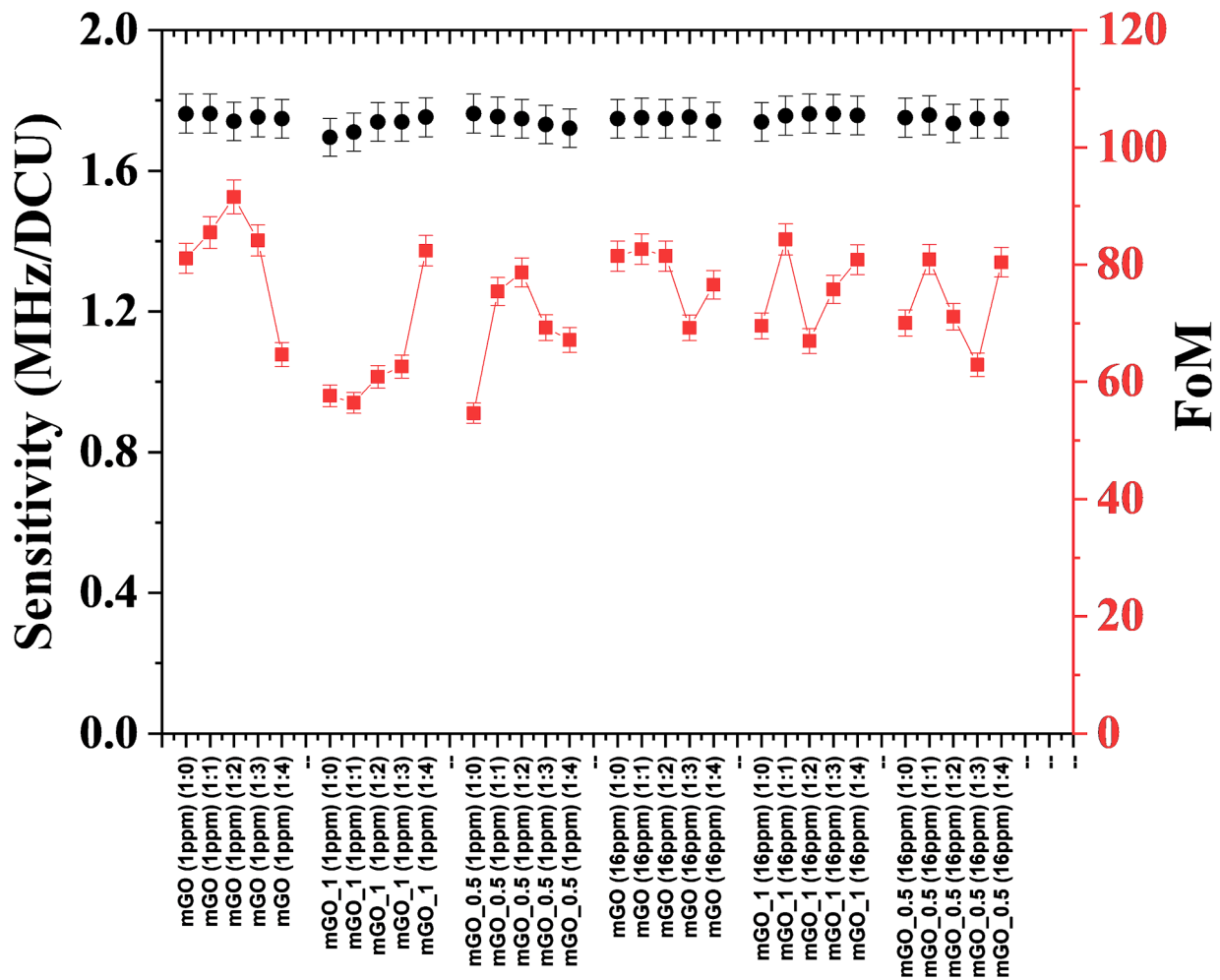


Fig. 9. Bar chart of sensitivity and figure-of-merit (FoM) of 1 ppm solution of mGO, mGO-1 and mGO-0.5 for different concentrations of BSA as depicted in Table 2.

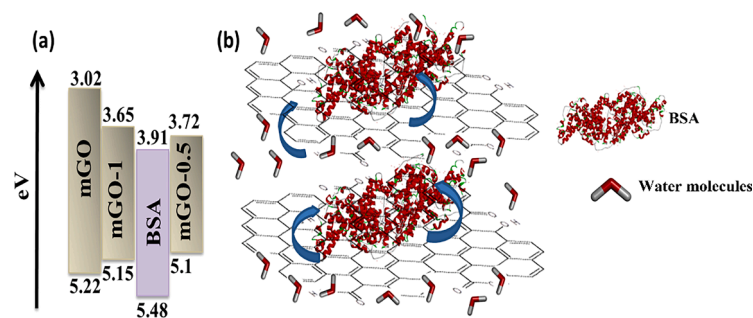


Fig. 10. (a) Energy level diagram of mGO, mGO-1 and mGO-0.5 with BSA, (b) Schematic for interaction of GO with BSA molecules in aqueous solution.

3.3. Single-split resonator sensor study for sensitivity towards BSA

Fig. 6a illustrates that the fabricated device showing the resonant frequency of 481MHz in air. Further the simulated SSR shows the resonant frequency at 500MHz for device which is within the acceptable limit. Furthermore, for the fabricated and simulated device, the corresponding S11 amplitude for the obtained resonant frequency peak is approximately -20.6dB and -41.82dB , respectively. Even though there is a difference between the two sets of data, the simulated resonant frequency peak shows high depth with respect to the experimentally measured resonant frequency peak. And it also shows the frequency shift between the simulated and measured resonant frequencies. These depth

and shift occurred due to some possible causes, including ohmic losses, polarization losses, and dielectric losses at the resistive-splitting [6]. The corresponding Q-factor, which defines the sharpness of the resonant peak, has been calculated for simulated and fabricated device to be 81.03 and 46.16, respectively. It is defined as the ratio of the observed resonant frequency (f_r) to the full-width and half-maximum (FWHM) of the same peak. With a high Q-factor, the biosensor may be more suitable for use as a highly sensitive and label-free biosensor for the detection of synthesis material at different concentrations of BSA [35,36]. Fig. 6b shows the variation to ϵ_{eff} of the material with respect to the frequency (2.77 with respect to resonant frequency 500MHz).

The distribution of the electric field of the SSR device is illustrated in

Fig. 6c. Compared to the rest of the structure, the electric field around the resistive-type split gap is strong at resonant frequency. The electric field reaches its maximum at approximately $5.36 \times 10^4 \text{ V/m}$. The red arrows show the propagation direction of electric field vectors. The electric field distributions for the air are mostly concentrated near the resonator surface. The sensing process is illustrated schematically in Fig. 7. The fabricated device has been characterized on a vector network analyzer (VNA). The permittivity of adjacent materials influences the measured resonant frequency of the fabricated SSR sample. Therefore, the change or shift in resonant frequency is attributed to a change in the permittivity of the materials that are surrounding the proposed sample.

The S-parameters of the bare resonator was characterized in the frequency range of 250MHz to 600MHz. This was done in order to assess the analytical performance of the biosensor. The frequency shift, due to increasing BSA concentration in GO solution is depicted in Fig. 7. The resonant frequency and S11 parameter of the bare resonator is measured in presence of the air (481MHz and -20.6dB , respectively). The resonant frequency and S11 parameter have been changed when a drop of $10\mu\text{L}$ of DI water is added at the resistive splitting. This change took place as a result of the electromagnetic wave that originated from the input port of the VNA, propagating through the device and interacting with the dielectric constant ($\sim 75-80$) of DI water at the resistive splitting. Subsequently, the loss became apparent in the propagating radiation i.e., the absorbance of electromagnetic wave has been occurred in DI water. The shift in the resonant frequencies due to the presence of testing materials (TMs, GO & GO:BSA) can be described in terms of the effective permittivities, as shown by the following equation (9)

$$f_{r,TMs} = f_{r,Ref} \cdot \sqrt{\frac{\epsilon_{eff,Ref}}{\epsilon_{eff,TMs}}} \quad (9)$$

where, $f_{r,Ref}$ and $\epsilon_{eff,Ref}$ are the resonant frequency and effective permittivity of resonator in present of reference material. $f_{r,TMs}$ and $\epsilon_{eff,TMs}$ are same in presence of TMs. If the ϵ_{eff} of air is ~ 1.0006 and $f_{r,Air}$ is 481MHz and $f_{r,TMs}$ of DI water is 354MHz then $\epsilon_{eff,TMs}$ of DI water is 75. The sensitivity of our sensor with respect to dielectric constant unit (DCU) is 1.71MHz/DCU.

Fig. 8 shows the resonance frequency positions of GO and increasing amount of BSA in GO, i.e., 0.5 ppm to 2.0 ppm (1:1, 1:2, 1:3, 1:4 respectively of 5 μM solutions) and Table 2 summarizes different parameters calculated. The sensitivity changes for each GO solution of 1 ppm and 16 ppm conc. with BSA addition. 1 ppm solutions show better shift in f_R and, therefore, change in sensitivity than 16 ppm as also observed in absorption studies. As can be seen in Fig. 9, 1 ppm mGO-1 shows sequential increase in sensitivity on increasing amount of BSA corroborating well with the absorption studies. FoM also increases from 57 to 82 and highest Q-factor calculated for mGO-1 is ~ 47 making it a good candidate as probe molecule. Also, 1 ppm mGO-0.5 shows sequential shift in frequency with increasing amount of BSA and therefore change in sensitivity and FoM. Other GO solutions, however show some overlap of frequency at different BSA concentrations.

4. Discussion

From the binding constant and % S data of absorption spectroscopy as well as SSR studies it is evident that 1 ppm solution of mGO-1 and mGO-0.5 provide good binding with BSA, at the same time effectively sensitive for all the concentrations. To understand the selectivity of GO towards proteins, we tested the GO material with ovalbumin protein and all showed very low or no sensitivity (Figure S6), indicating high sensitivity and selectivity of mGO towards BSA sensing. Gibb's free energy is calculated using the K_c in $\Delta G = -RT \ln K_c$ [37]. ΔG is calculated to be -40.7 and -39.3 kJ/M for mGO, -38.8 and -36.0 kJ/M for mGO-1 and -44.3 and -48.2 kJ/M for mGO-0.5, 1 ppm and 16 ppm solutions respectively. The values show spontaneous interaction

between GO and BSA. The energy level diagram related to band gap of mGO, mGO-1, mGO-0.5 and BSA is shown in Fig. 10a. Proteins have been exploited for applications in nanoscale device due to their small single molecular size, effective electron transport process capabilities and optical characteristics. BSA biomolecule has been utilized in miniature transistor devices by Elat et al. [38]. The band gap of free BSA is calculated to be ~ 1.5 eV and LUMO is deeply located at ~ 3.91 eV indicating high electron affinity in the presence of electron rich materials like GO [39]. Energy level alignment of mGO-1 and mGO-0.5 is providing sufficient driving force for charge-transfer from GO to BSA with good binding affinity. The suitable band gap of mGO-1 and mGO-0.5 also allows better sensitivity towards BSA. Dilution on the other hand plays role in sensitivity and 1 ppm solution due to better interaction interfaces allow better sensitivity for all concentration ranges of BSA (Fig. 10b). Table S1 shows the comparative performance of different ring resonator devices for biomolecules with the sensitivity and response range. Both, mGO-1 and mGO-0.5 in the current study show good sensitivity in the range of MHz region for quantitative detection of BSA, however mGO-1 shows best probe for quantitative estimation of BSA using SSR biosensor device.

5. Conclusion

We have demonstrated the sensitivity of 1 ppm solution of mGO-1 and mGO-0.5 with band gap of 1.5 and 1.35 eV respectively towards quantitative identification of BSA using both spectroscopy method and SSR based biosensor device. 1 ppm solution of mGO-1 is found to be the best probe for quantitative identification of BSA showing sequential increase in sensitivity as well as FoM on increasing amount of BSA. Our designed SSR device is working in megahertz region of frequency with simple design and requiring only a small amount of test sample. mGO-1 can be synthesized repeatably with same degree of functionalization and band gap and thus the performance is repeatable and reproducible.

CRediT authorship contribution statement

Rachana Kumar: Writing – original draft, Project administration, Writing – review & editing, Supervision. **Rashmi Gautam:** . **Ankit Singh:** . **Suyashi Singh:** Data curation. **Pramod Kumar:** .

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.cartre.2024.100367](https://doi.org/10.1016/j.cartre.2024.100367).

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